MIL-P-43045B(MR)
22 February 1974
SUPERSEDING
MIL-P-43045A(MR)
19 January 1968

MILITARY SPECIFICATION

PLASTIC POTTING AND IMPREGNATING MATERIAL, POLYBUTENE

1. SCOPE

1.1 This specification covers one type of polybutene potting and impregnating material.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-96 - Can, Metal, 28 Gage and Lighter.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-D-760 - Drum and Pail, Metal (5 and 16.64 Gallon).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specifid procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 9330

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American Society for Testing and Materials (ASTM) Standards:

D 70-72 - Specific Gravity of Semi-Solid Bituminous Materials.

D 88-56 (1968) - Saybolt Viscosity.

D 92-72 - Flash and Fire Points by Cleveland Open Cup.

D 150-70 - A. C. Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.

D 257-66 (1972)- D. C. Resistance of Conductance of Insulating Materials.

D 445-72 - Kinematic Viscosity of Transparent and Opaque Liquids and the Calculation of Dynamic Viscosity.

D 664-58 (1968) - Neutralization Number by Potentiometric Titration.

D 924-65 (1969) - Power Factor and Dielectric Constant of Electrical Insulating Liquids.

D 972-56 (1968) - Evaporation Loss of Lubricating Grease and Oils.

D 974-64 (1973) - Neutralization Number by Color-Indicator Titration.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street N. W. Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The material shall consist of butene polymers which are free of additives (see 4.2.1).

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3.2 Property values. The polybutene shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.

Table I. Property values

Property	Maximum	Minimum
Viscosity at 99°C (210°F) Saybolt Universal seconds	3200	2800
Specific gravity at 25°C/25°C (77°F/77°F)	0.905	-
Flash point, open cup, °C	-	199
°F	-	390
Loss on heating, percent	2.0	-
Total acid number, milligrams of potassium hydroxide per gram	0.10	-
Volume resistivity at 500 volts (1 minute), ohm-centimeters, at 23° ± 1°C (73.4 ± 1.8°F)	-	1×10^{15}
Dielectric constant, at 1 megacycle per second, at 23° ± 1°C (73.4° ± 1.8°F)	2.30	-
Dissipation factor, (tan delta), at 1 megacycle per second, at 23° ± 1°C (73.4° ± 1.8°F)	0.0015	-

3.3 Color. The color of the material shall be natural, which is clear and colorless.

3.4 Workmanship. The material shall appear uniform. The material shall be free from dirt, metallic particles, and any foreign material, as determined visually.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified. In the event of conflict, this specification will govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 50 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table II shall be approximately one pound.

Table II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and Workmanship	Not clear and colorless
	Material not uniform
	Not clean, presence of foreign material

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4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with table III, to determine that packaging, packing and marking comply with requirements specified in section 5. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table III. Examination of preparation for delivery.

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers
Quantity of Material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect incomplete of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

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4.2.3 Testing. The potting and impregnating material shall be tested for the characteristics listed in table I in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 50 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be the averaged results of the two tests per sample unit.

4.3 Test methods

4.3.1 Conditions. Test temperature shall be in accordance with the applicable procedure specified in table IV and 50 ± 5 percent relative humidity. Prior to testing, the sample shall be conditioned at the test temperature for a minimum of 4 hours.

4.3.2 Tests. Testing shall be in accordance with the methods specified in table IV, as applicable.

Table IV. Test methods

Test	Number of tests per sample unit	ASTM Method
Viscosity	2	D 445-72 or D 88-56 (1968)
Specific gravity	2	D 70-72
Flash point	2	D 92-72
Loss on heating	2	D 972-56 (1968) Except reduce time of air passage from 22 to 10 hours at 99° ± 2°C (210° ± 3.6°F)
Total acid number[1/]	2	D 974-64 (1973) or D 664-58 (1968)
Volume resistivity	2	D 257-66 (1972)
Dielectric constant	2	D 924-65 (1969)
Dissipation factor	2	D 150-70 or D 924-65 (1969)

[1/] Use 0.10 normal potassium hydroxide (KOH)

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5. PREPARATION FOR DELIVERY

Application. The requirements of Section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packaging. Packaging shall be in the quantities and at the level specified (see 6.2).

5.1.1 Level A. The polybutene material shall be furnished in 1 pint or 1 gallon cans, 5 gallon pails, or 55 gallon drums as specified by the procuring activity (see 6.2).

5.1.1.1 Cans. Cans shall conform to type V, class 2 of PPP-C-96. Exterior plan B coating and side seam stripping shall be required.

5.1.1.2 Five gallon pails. The 5 gallon pails shall conform to type II of PPP-D-760.

5.1.1.3 Fifty-five gallon drums. The 55-gallon drums shall conform to type III or type IV of PPP-D-729 in quantities of 400 pounds, maximum.

5.1.2 Level C. Packaging shall be in cans, pails, or drums and shall provide adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, cans shall be arranged and packed for overseas shipment in accordance with the appendix to PPP-C-96. Gross weight shall not exceed a maximum of 200 pounds. Five gallon and 55-gallon containers will not require packing.

5.2.2 Level B. Unless otherwise specified, cans shall be arranged and packed for domestic shipment in accordance with the appendix to PPP-C-96. Five and 55-gallon containers will not require packing.

5.2.3 Level C. Packaged cans shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The polybutene material covered by this specification is intended primarily for use in potting and impregnating compounds.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of container for polybutene and applicable levels of packaging and packing (see 5.1 and 5.2).

Custodian:
Army-MR

Preparing Activity:
Army-MR

(Project No. 9330-A418)

Review
Army-EL, MI, WC

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: MIL-P-43045B(MR), Plastic Potting and Impregnating Material, Polybutene

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

DD, FORM, APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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